Micro I, class 4

We discussed a classical *Envelope Theorem*, which gives conditions under which a value function is differentiable and asserts what the derivative equals. We used it to confirm *Roy's Identity*,

$$d_i(p, w) = -\frac{\partial V/\partial p_i}{\partial V/\partial w}(p, w).$$

As mentioned, this identity gives a much simpler way to calculate demands than starting from a utility representation.

But it only holds for *indirect utility function*, not any old function of prices and wealth. How do we know whether a function of prices and wealth *is* an indirect utility function? Under our maintained assumption of a continuous and LNS preference representation, we agreed that it must be *nonincreasing in* p, and *strictly increasing in* w; *homogenous of degree 0* on its domain; and we proved together that it is *quasiconvex* on its domain. In words, the last says that a consumer prefers the *best* of two price-wealth pairs to an *average* of the two price-wealth pairs, a preference for *extremes* in budget sets, not averages. I argued that such a preference can be explained by *substitution effects* in consumption: when relative prices change, consumers can react by substituting towards relatively cheaper goods; when budget sets are averaged, such substitution possibilities are ruled out.

But to derive the testable implications of consumer theory, it is easier to be a little more indirect than the indirect utility function. We will instead work with a function that turns out to be the inverse of the indirect utility (for each fixed price), the *expenditure function.* It gives the minimum expenditure at prices p to be at least as well as the indifference set indexed by the utility number, $\bar{u}$. The set of solutions to this (EM) problem at $(p, \bar{u})$ is denoted $h(p, \bar{u})$ and called either the *Hicksian* or *compensated* demand.

The plan for out next couple of meetings are summarized in my notes entitled *Demand Properties: Summary.* I encourage you to read through it for an overview. We start with the Milgrom-Segal Envelope Theorem (on page 2 of my notes on the Envelope Theorem).